

The background image shows a large, dark, cylindrical submarine cable running along the seabed. The cable is supported by a series of vertical pillars that form a bridge-like structure extending into the distance. The water is a deep blue, and the sky is filled with soft, white clouds. The seabed is visible in the foreground, showing rocks and sand.

SECURITY AND RESILIENCE OF SUBSEA CABLE INFRASTRUCTURE AND MARITIME SAFETY ENHANCEMENT

The security of submarine cables is a growing global concern. These communication and energy supply networks are essential for international connectivity and the functioning of modern society. Recent incidents in the Baltic Sea highlight the vulnerability of these critical infrastructures.

The European Commission's Recommendation of 26.2.2024 on the security and resilience of submarine cable infrastructure presents a set of actions at the national and EU level aimed at improving submarine cable security and resilience through better coordination across the EU, both in terms of governance and funding.



It outlines a series of actions at both national and EU levels. The aim is to enhance the security and resilience of submarine cable systems, which are vital for global communications and data transfer. These actions are designed to address the challenges faced by submarine cable infrastructures and to ensure their continued operation and reliability in the face of potential threats:

Improving Coordination:

Establishing better coordination across the EU to manage and protect submarine cable infrastructures.

Enhancing Governance:

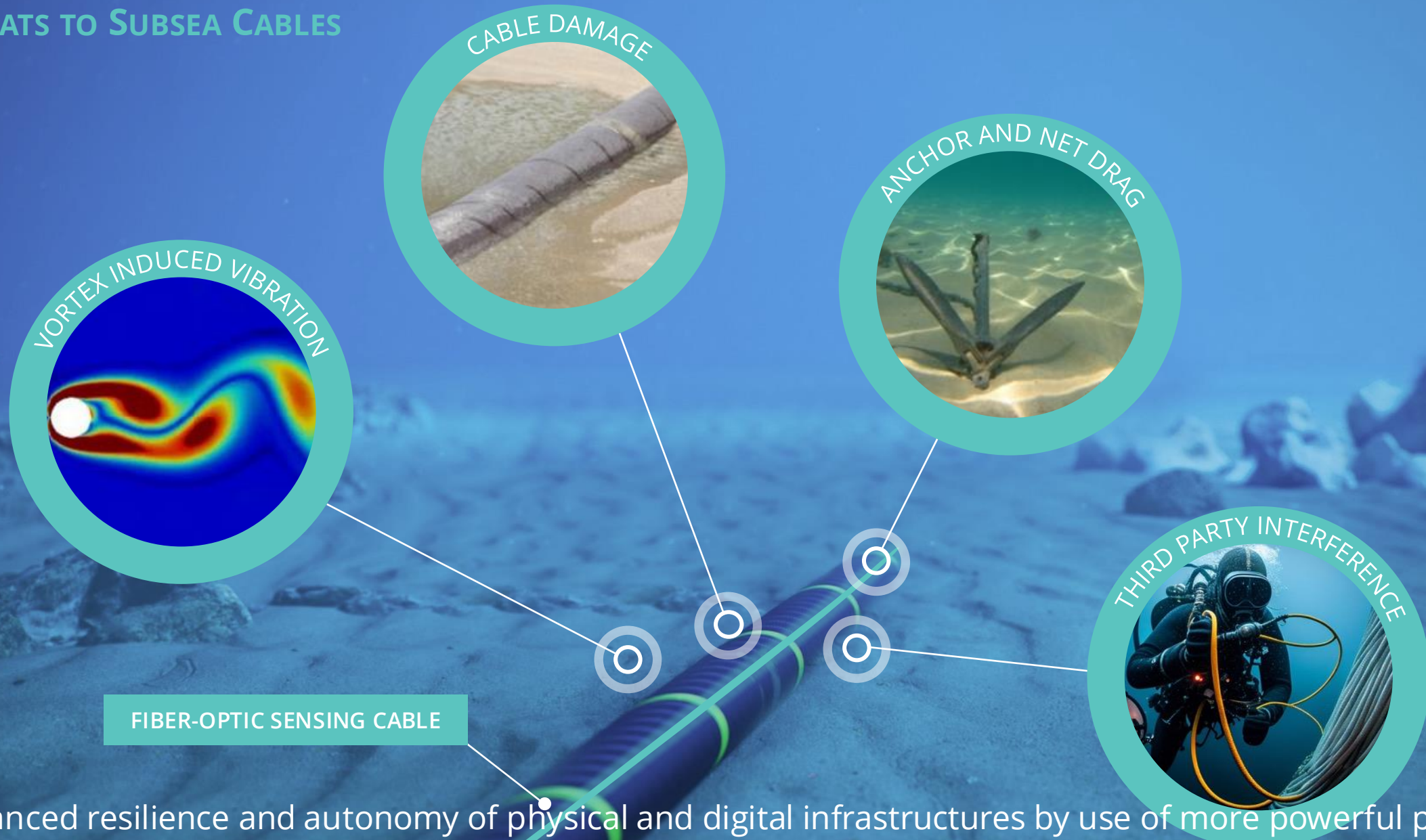
Strengthening governance structures to ensure the security and resilience of these critical assets.

Increasing Funding:

Allocating appropriate funding to support the necessary measures to safeguard submarine cable systems.

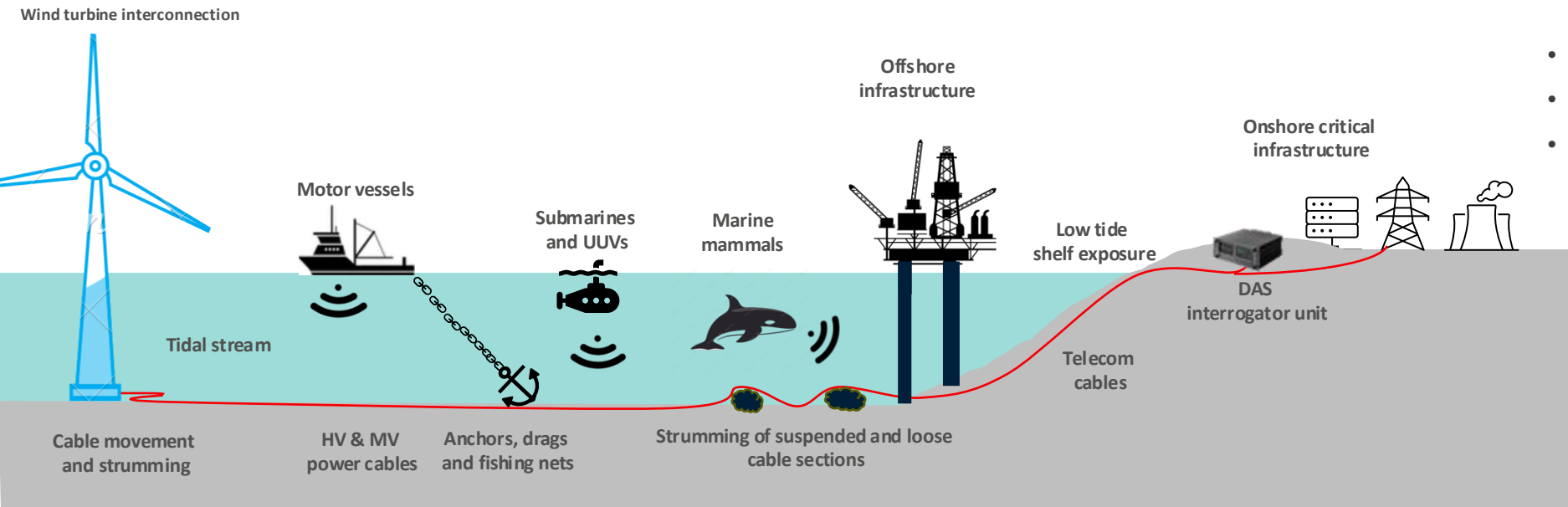


THREATS TO SUBSEA CABLES

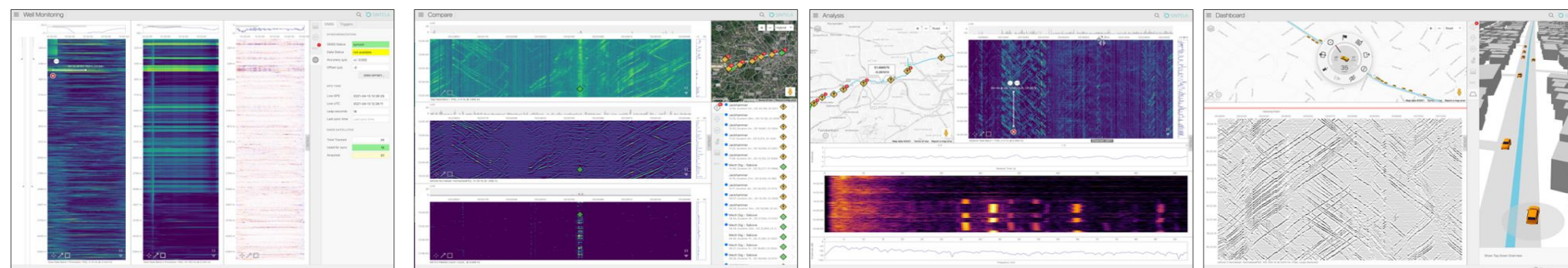


Enhanced resilience and autonomy of physical and digital infrastructures by use of more powerful prevention, preparedness and response, a better understanding of related human, societal and technological aspects, and the development of cutting-edge capabilities for critical infrastructure operators.

SOLUTION: ADVANCED DISTRIBUTED ACOUSTIC SENSING (DAS) TECHNOLOGY PROTECTING SUBSEA CABLES



- Using dark FO fibre in existing cables
- Sensing cable deployed on the seabed
- Detection of waterborne activities of interest such as:
 - Anchoring, net dragging
 - Third party interference
 - Motorboats (e.g. if AIS is off)
 - Unmanned underwater vehicles
 - Diver delivery vehicles
 - Cable condition monitoring
 - Shallow water, low tide seabed monitoring

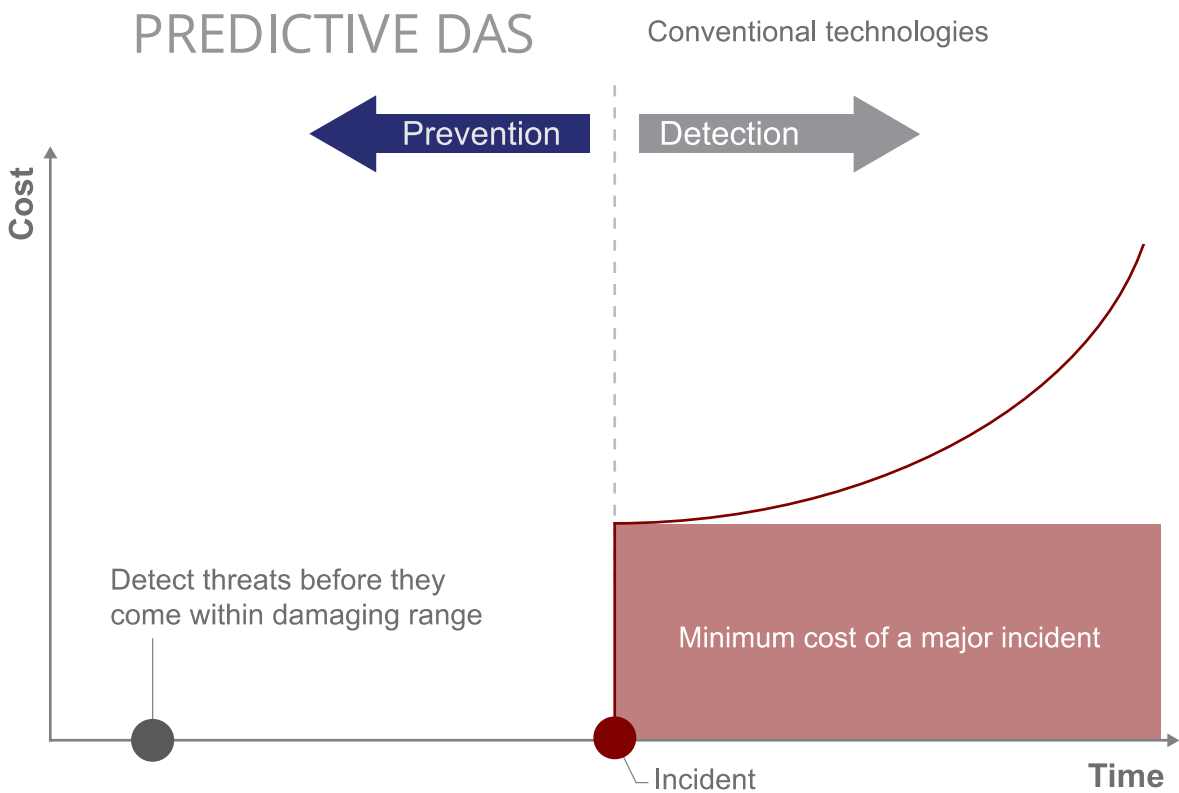


Example screenshots from various tabs of the Web-based User Interface

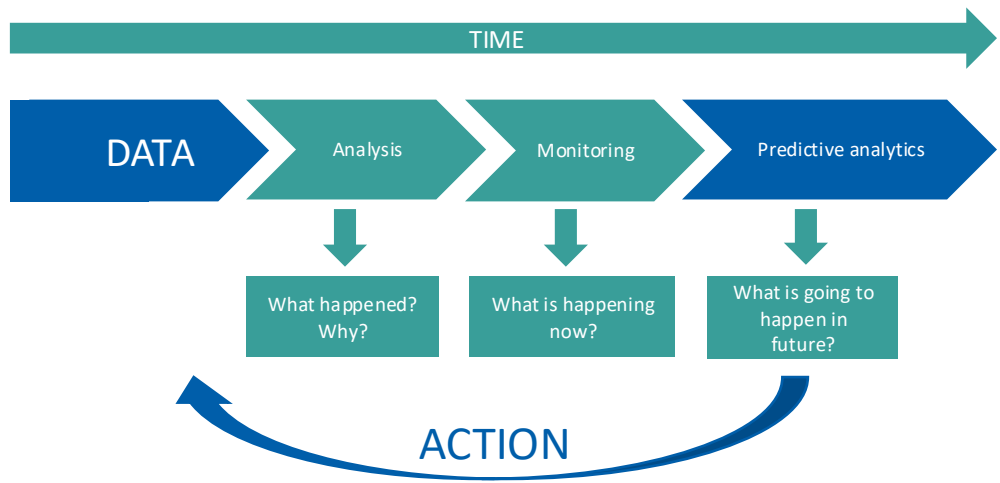


KEY ADVANTAGE – PREVENTION USING AI AND ALGORITHMS FOR BIG DATA ANALYTICS

The technology that can provide information before a damaging incident occurs



PREDICTIVE DAS DATA ANALYTICS:



Enhancing the resilience of large-scale interconnected subsea cable systems and their operators against complex attacks, natural and human-made disasters, and the effects of climate change.

Upgraded systems to enhance operator resilience and safeguard critical infrastructure, enabling rapid, effective, safe, and secure responses with minimal human intervention to complex threats and challenges, while improving risk assessment, ensuring resilience and strategic autonomy of European subsea infrastructure.

FOSMARIN SENSE

FOSMARIN PREDICT

FOSMARIN COMMAND

SINTELA
DAS Technology Provider

melcano
System Integration



DAS Interrogator

real-time DAS data:

- strain rate
- spectral energy
- higher level information

FORTH
FOUNDATION FOR RESEARCH AND TECHNOLOGY - HELLOS
Interrogator and Seismic Algorithms

training data

AIT
AUSTRIAN INSTITUTE OF TECHNOLOGY
Algorithms

AIS Tracking Data

data fusion

ADDRESS
Command and Control Interface



User Interface for Operators and Authorities



Data Storage

FORTH

Test Sites

canalink

PLACAN
Plataforma Operativa de Canarias



Requirements Analysis
Workshops
Dissemination
Training

End-Users

GlobalConnect

SHEFA

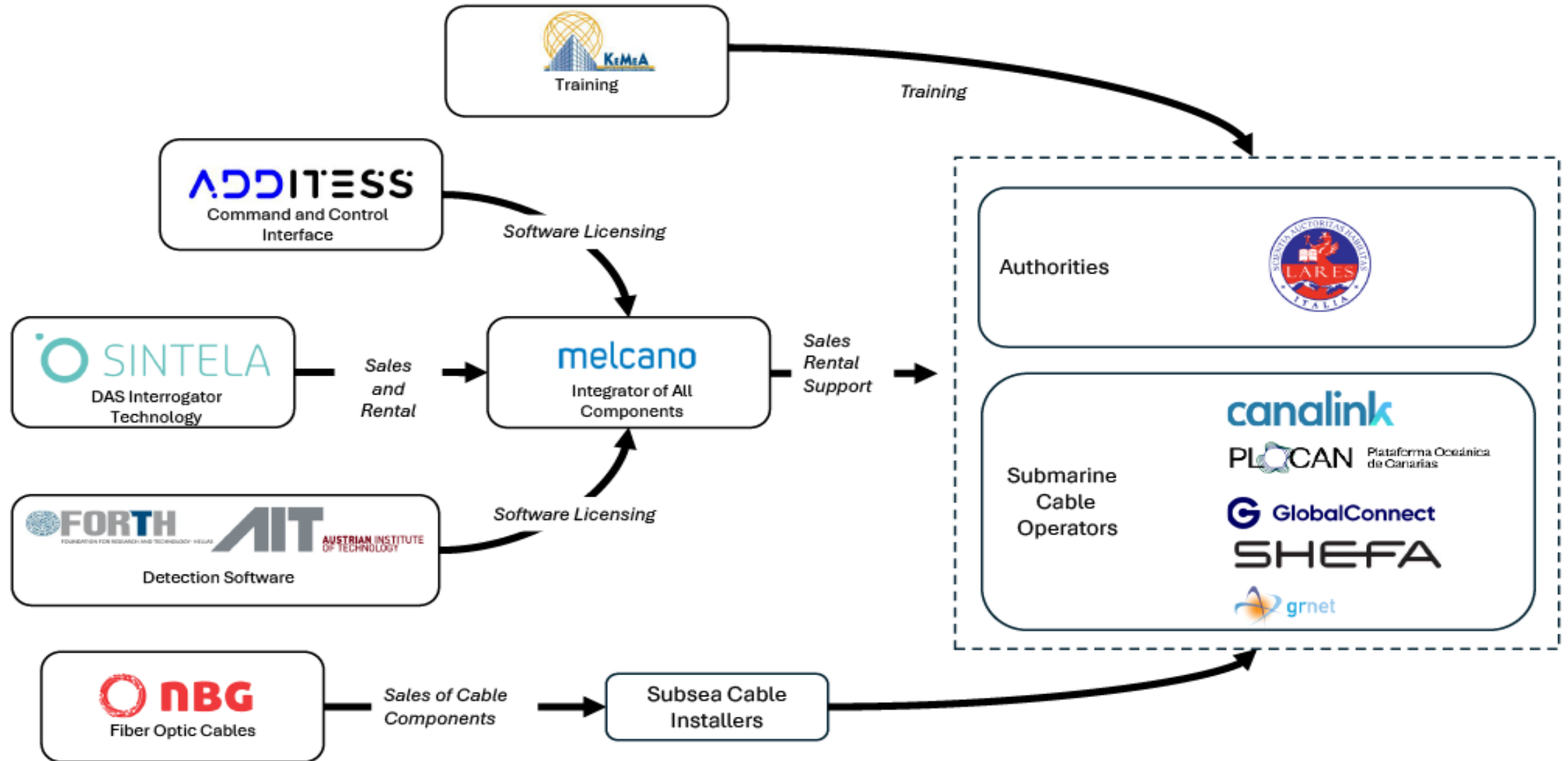


NBG
special purpose fibre cable technology

Subsea infrastructure with fiber optic cable



Consortium Structure





Fibre Optic Sensing for subMARINe resilience

CONSORTIUM:

Data Science:

AIT
FORTH

Austria
Greece
UK

DAS technology:

SINTELA
MELCANO
ADDITESS

Austria
Cyprus

System Integrators:

NBG
CANALINK
PLOCAN
Global Connect

Austria
Spain
Spain
Sweden
Norway
Germany
Denmark
Finland
Denmark
Greece
Italia
Greece

FO cable technology:

Operators:

SHEFA
Grnet
LARES
KeMeA

Authorities:

Training:

